

## 2023 Calculus II    Additional Exercise (6)

1. Exercise 8 in page 35 from Calculus Supplement Homework.
2. Exercise 5 in page 35 from Calculus Supplement Homework.
3. Exercise 18 in page 36 from Calculus Supplement Homework.
4. Suppose a line  $l$  passes through  $P(1, 2, 1)$  and perpendicular to the line  $x = 1 + 3t$ ,  $y = 2t$ ,  $z = -1 + t$ . If  $l$  intersect with the line  $x = 2s$ ,  $y = s$ ,  $z = -s$ , find the equation of  $l$ .
5. Given the lines  $L_1$ :  $x = 1 + 4t$ ,  $y = 1 + 2t$ ,  $z = -3 + 4t$  and  $L_2$ :  $x = 3 + 2s$ ,  $y = 2 + s$ ,  $z = -2 + 2s$ . Can  $L_1$  and  $L_2$  determine only a plane  $M$ ? If so, find the equation of  $M$ .